

# Case 14: Two-Dimensional Spatial-Resolution Sensitivity

## 1 Verification purpose

**Case ID:** CASE14\_SRS

This case is a negative verification test for mesh dependence in the current multiple-discrete-cell treatment of structures. A physical 10 m-by-10 m house may occupy one raster cell, several independent cells, one oversized cell, or disappear under categorical resampling as the mesh changes. Because those cells are not assembled into one conserved structure object, changing resolution alters represented heat release, firebrand generation, deposition, and structure-to-structure ignition.

The test asks whether all 15 current-fingerprint simulations produce an evaluable community rate of spread (ROS), and whether the fitted values remain within the predeclared mesh-independence limits relative to the run-derived 10-m anchor. Detecting the known resolution sensitivity is useful diagnostic evidence, but it is a *verification failure*, not a pass.

## 2 Mathematical formulation and numerical implementation

### 2.1 Mode-resampled square array

The physical layout is a periodic array of square structures of side  $a = 10$  m, separated in both coordinates by  $d = 10$  m. Define the one-dimensional structural indicator

$$I(s) = \begin{cases} 1, & s \bmod (a + d) < a, \\ 0, & \text{otherwise.} \end{cases}$$

For grid interval  $C_i = [i\Delta, (i + 1)\Delta]$ , preprocessing calculates

$$f_i = \frac{1}{\Delta} \int_{C_i} I(s) ds, \quad b_i = \mathbf{1}(f_i \geq 1/2), \quad B_{ij} = b_i b_j.$$

This coordinate-wise categorical-mode construction reproduces the reported geometries. At  $\Delta = 7$  m, effective side lengths are 7 or 14 m and footprints range from 49 to 196 m<sup>2</sup>. At  $\Delta = 10$  m, footprint and gap are exact. At  $\Delta = 30$  m, one represented structure cell has area 900 m<sup>2</sup> and represented structures are separated by 30 m. The 20- and 40-m maps eliminate represented firebreaks; this is an intended resampling diagnostic, not a preprocessing failure.

### 2.2 Cell-local firebrand treatment

A represented structural cell uses a design-fire curve  $q''(t)$ . The curve has a 300-s linear growth phase, a one-hour plateau at 400 kW m<sup>-2</sup>, and a 300-s decay. Under the current multiple-discrete-cell treatment,

$$\text{HRR}_{ij}(t) = q''(t)\Delta^2, \quad \dot{N}_{ij}(t) = GR \frac{\text{HRR}_{ij}(t)}{1000}, \quad GR = 10 \text{ pcs MW}^{-1} \text{ s}^{-1}.$$

Changing  $\Delta$  therefore changes both represented geometry and the HRR of each independent emitter. Firebrands use the structural empirical flight-distance distribution, Eulerian accumulation, and physical ignition. Non-structure cells use nonburnable fuel model 93, so vegetative surface spread is absent and every downwind ignition must be caused by deposited firebrands. The global NO\_SURFACE\_FIRE switch remains true to suppress vegetative surface-fire propagation. The current source independently

refreshes initialized model-14 building transient HRR under this setting. The configured time-step scaling is documented by Qin et al. [2]:

$$\Delta t = \text{CFL} \frac{\Delta}{U_{\text{wind}}}, \quad \text{CFL} = 1, \quad U_{\text{wind}} = 17.9 \text{ m s}^{-1} (40 \text{ mph}).$$

The fractional step is required to hold this spatial sweep at fixed wind CFL. Preprocessing sets each generated stop to  $\lceil 150000/\Delta t \rceil \Delta t$ , then records the requested stop, aligned stop, and integer step count in the manifest.

For each represented downwind structure column  $s$ , postprocessing takes the median positive finite arrival time  $t_s$ . At least three ignited columns are required. Mean community ROS is the least-squares slope

$$ROS_{\Delta} = \frac{\sum_s (t_s - \bar{t})(x_s - \bar{x})}{\sum_s (t_s - \bar{t})^2}.$$

Because the 10-m grid exactly resolves both physical length scales, its fitted ELMFIRE result is the comparison anchor,

$$ROS_{10} = ROS_{\Delta=10 \text{ m}}.$$

The dashed reference line and the 10-m ELMFIRE point therefore use the same calculated value; no rounded external estimate is substituted.

### 3 Derivation of expected behavior

A resolution-independent structure treatment would conserve each house's physical footprint, integrated heat release, firebrand source, accumulated load, and ignition state. Subject to numerical error, it would give  $ROS_{\Delta} \approx ROS_{10}$  for every resolution and enough downstream ignitions to fit all 15 ROS values.

The current cell-local representation does not conserve those quantities. Fine meshes fragment houses into multiple emitters, intermediate meshes alter footprints and gaps, and some coarse meshes merge structures or remove firebreaks. The expected diagnostic result is therefore a nonuniform ROS curve, including possible failure to propagate at some resolutions. That observation exposes the defect; against the mesh-independence requirement, its correct overall status is FAIL.

The period of the one-dimensional physical indicator is  $a + d = 10 + 10 = 20 \text{ m}$ . The key raster substitutions are therefore:

| $\Delta \text{ (m)}$ | mode-resampled consequence                  | component-area reference                        |
|----------------------|---------------------------------------------|-------------------------------------------------|
| 7                    | one or two selected cells per coordinate    | $7^2 = 49$ to $(2 \cdot 7)^2 = 196 \text{ m}^2$ |
| 10                   | one structure cell followed by one gap cell | $10^2 = 100 \text{ m}^2$                        |
| 30                   | one selected cell per represented structure | $30^2 = 900 \text{ m}^2$                        |

At  $\Delta = 20 \text{ m}$ , each cell contains 10 m of structure and 10 m of gap, so  $f_i = 0.5$  and the  $f_i \geq 0.5$  rule selects every cell; the represented firebreak vanishes. The same 50-percent selection occurs for the 40-m map. These are reproducible consequences of the generated mode rule, not inferred from the output fire spread.

With  $U = 40(0.44704) = 17.8816 \text{ m s}^{-1}$ , each generated step is

$$\Delta t = \Delta/U; \quad \Delta t_7 = 0.391463, \quad \Delta t_{10} = 0.559234, \quad \Delta t_{30} = 1.677702 \text{ s}.$$

A truly resolution-independent treatment would give all 15 finite fits,  $f_{\text{fit}} = 15/15 = 1$ ,  $ROS_{\Delta} = ROS_{10}$ ,  $CV_R = 0$ , and  $D_{\text{max}} = 0$ . The current native cell treatment is expected instead to produce missing or nonuniform fits because source HRR scales as  $\Delta^2$  and the represented geometry changes.

The predeclared mesh-independence requirements permit  $CV_R \leq 0.25$  and  $D_{\max} \leq 0.50$ ; exceeding either limit or failing any of the 15 fits yields the expected diagnostic FAIL.

The primary publication provides the independent propagation oracle used only for strict stalled-final compatibility: Figure 9 reports zero ROS at 1, 3, 7, 13, 17, 23, 25, and 27 m, and finite propagation at 5, 10, 15, 20, 30, 40, and 50 m [2]. An early stalled-final dump may therefore use the no-propagation exception only for the first set, and only after the meteorology-interval reconstruction and timestep-grid checks also pass. This oracle does not replace the current-run metrics or authorize accepting an arbitrary overrun timestamp.

## 4 Verification metrics and acceptance criteria

Preprocessing passes only when all 15 resolutions are generated, the exact 10-m map has no footprint or gap error, the 7-m footprint range is 49–196 m<sup>2</sup>, and the median 30-m footprint is 900 m<sup>2</sup>. These checks establish that runtime comparison uses the intended resolution-dependent rasterizations.

A complete runtime decision additionally requires every current-fingerprint variant and a finite 10-m ROS anchor. Let  $N_{\text{fit}}$  be the number of resolutions with at least three ignited downwind structure columns. ROS extraction completeness is

$$f_{\text{fit}} = \frac{N_{\text{fit}}}{15},$$

and must equal one. For the finite fitted values, mesh dependence is measured by the coefficient of variation

$$CV_R = \frac{\text{std}_{\Delta}(ROS_{\Delta})}{\text{mean}_{\Delta}(ROS_{\Delta})}$$

and by the largest deviation from the exact-geometry anchor,

$$D_{\max} = \max_{\Delta} \frac{|ROS_{\Delta} - ROS_{10}|}{ROS_{10}}.$$

The permissive engineering limits are  $CV_R \leq 0.25$  and  $D_{\max} \leq 0.50$ . Missing fits are handled by  $f_{\text{fit}}$ ; they are not silently discarded and counted as convergence. The overall decision is the conjunction of every geometry, runtime, anchor, fit-completeness, and mesh-independence check.

### 4.1 Justification of metric quantification

The geometry limits are exact consequences of rasterization. At  $\Delta = 10$  m, the structure side and gap are both exactly one cell, so the 1% footprint and separation allowances cover only file precision and connected-component bookkeeping. At 7 m, a represented component can contain one to four cells, giving areas from  $7^2 = 49$  to  $4 \times 7^2 = 196$  m<sup>2</sup>. At 30 m, a one-cell represented structure has area  $30^2 = 900$  m<sup>2</sup>. These checks establish that the intended mesh-dependent maps were generated; they are not claims of geometric accuracy.

For the behavioral requirement,  $f_{\text{fit}} = 1$  is non-negotiable because mesh independence cannot be inferred after silently discarding resolutions that fail to propagate. The  $CV_R \leq 0.25$  and  $D_{\max} \leq 0.50$  limits are intentionally permissive engineering bounds, substantially larger than ordinary front-fit quantization. Even with those allowances, a method with factor-of-two variation or missing fits fails, which is the defect this case is designed to expose [2]. The 10-m reference is the actual  $ROS_{10}$  extracted by the same postprocessor, so its plotted reference and ELMFIRE point must coincide by construction; it is not an independent nominal value. Current fingerprints and all 15 runs are required to prevent stale or selectively complete outputs from weakening the expected FAIL diagnosis.

## 4.2 Predeclared acceptance contract

Table 1 summarizes the metrics fixed before inspection of the calculated results. Numerical values and component statuses are reported separately in Section 7.

Table 1: Predeclared verification metrics and acceptance criteria.

| Metric                      | Output, region, and aggregation                                                 | Acceptance criterion       |
|-----------------------------|---------------------------------------------------------------------------------|----------------------------|
| Prepared variants           | Generated manifest count divided by the 15 requested grid spacings.             | 100%                       |
| 10-m footprint error        | Maximum relative error of represented component areas from 100 m <sup>2</sup> . | $\leq 0.01$                |
| 10-m separation error       | Maximum relative gap-width error from 10 m.                                     | $\leq 0.01$                |
| 7-m component areas         | Minimum and maximum represented connected-component areas.                      | 49–196 m <sup>2</sup>      |
| 30-m footprint error        | Relative error of median represented footprint from 900 m <sup>2</sup> .        | $\leq 0.01$                |
| Runtime completeness        | Current-fingerprint final TOA output and successful termination.                | 15 of 15 variants          |
| 10-m ROS anchor             | Structure-column fit from the current 10-m TOA raster.                          | Finite $ROS_{10}$          |
| ROS extraction completeness | Resolutions with at least three ignited downwind columns.                       | $f_{\text{fit}} = 1$       |
| ROS mesh-dependence CV      | Coefficient of variation over finite fitted ROS values.                         | $CV_R \leq 0.25$           |
| Maximum anchor deviation    | Maximum finite-grid relative deviation from $ROS_{10}$ .                        | $D_{\text{max}} \leq 0.50$ |
| Overall decision            | Every geometry, completeness, and ROS component is conjunctive.                 | PASS only if all pass      |

## 5 Simulation configuration and predicted outcome

Figure 1 records the prepared spatial inputs for dx10. The postprocessor reads these panels from the variant-local GeoTIFF files, removes the two-cell numerical buffer, and retains map coordinates in metres. The left panel shows the most informative fuel or structure layout available for the case; the right panel shows the cells selected by the non-positive initial level-set field, making a localized ignition or firebrand-emission source explicit.

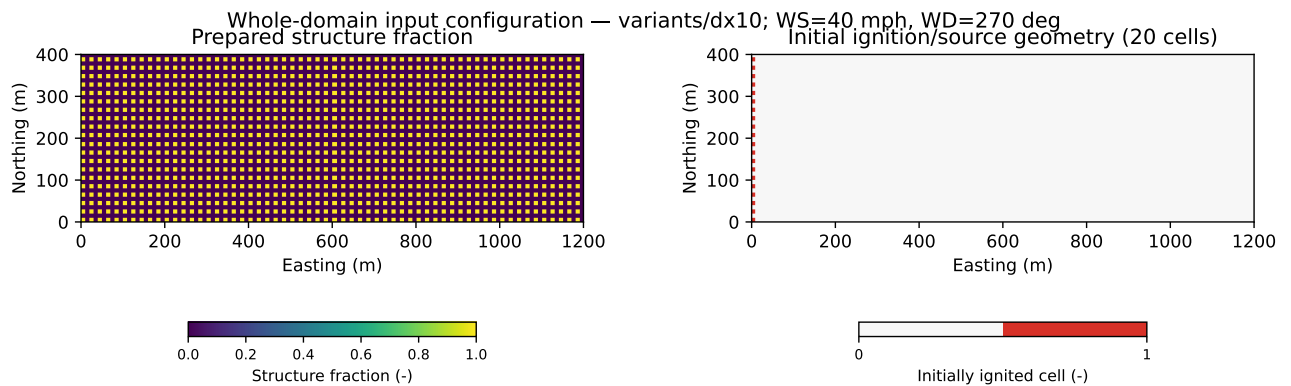


Figure 1: Whole-domain prepared input configuration for the representative variant. The physical domain is shown without the numerical halo; coordinates, configured wind values, fuel or structure layout, and initial ignition/source geometry are identified in the panels.

Table 2: Configuration of the two-dimensional resolution sweep.

| Parameter                 | Configured value                                       | Units                              | Role                                            |
|---------------------------|--------------------------------------------------------|------------------------------------|-------------------------------------------------|
| Physical domain           | 1200 × 400, flat                                       | m                                  | permits many downwind ignitions                 |
| Structure side / gap      | 10 / 10                                                | m                                  | exact physical scales at the 10-m grid          |
| Initial ignition          | complete first structure column                        | –                                  | uniform upwind source across all structure rows |
| Grid spacings             | 1, 3, 5, 7, 10, 13, 15, 17, 20, 23, 25, 27, 30, 40, 50 | m                                  | exposes raster and cell-local dependence        |
| Numerical halo            | 2 on every side                                        | cells                              | excluded from physical geometry                 |
| Wind                      | 40 from 270 degrees                                    | mph                                | constant west-to-east transport                 |
| Time step                 | $\Delta/U_{\text{wind}}$                               | s                                  | holds wind CFL at one                           |
| Design-fire peak          | 400                                                    | $\text{kW m}^{-2}$                 | structural HRRPUA input                         |
| Generation rate           | 10                                                     | $\text{pcs MW}^{-1} \text{s}^{-1}$ | cell-local building source                      |
| Transport / accumulation  | empirical / Eulerian                                   | –                                  | current structural spotting path                |
| Ignition delays           | 42 plus 300                                            | s                                  | small-flame and development stages              |
| Vegetative surface spread | excluded by fuel 93                                    | –                                  | isolates firebrand-driven propagation           |
| Stop time                 | at least 150000                                        | s                                  | rounded up to an exact timestep multiple        |
| Final output              | one TOA per variant                                    | –                                  | source for structure-column ROS fits            |

The dashed reference line is predicted to intersect the 10-m ELMFIRE point exactly because both are  $ROS_{10}$ . A mesh-independent implementation would cluster all other finite ROS points around this line and provide a fit at every resolution. The current independent-cell treatment is instead expected to produce missing fits and large variations, causing the behavioral verification to fail.

## 5.1 Preprocessing, execution, and postprocessing

`scripts/preprocess.py` places customizable constants at its beginning, constructs the periodic map analytically, assigns persistent structure IDs, writes co-registered GeoTIFF inputs and case-local fuel tables, and records a SHA-256 fingerprint for every variant. It also writes `outputs/geometry_summary.csv`, making the map checks independently auditable.

`run_case.sh` is executable within the case directory and follows the same linear workflow as the other manifest cases. Configure `ELMFIRE_BIN` before launching it; the runner executes all 15 variants sequentially, keeps logs and fingerprints variant-local, postprocesses the outputs, and compiles the report. Because the sweep is long, execution should be scheduled only when adequate compute time is available.

`scripts/postprocess.py` rejects stale or incomplete outputs, reads the final time-of-arrival raster, groups cells by structure, fits ROS, calculates all metrics above, and produces the report figure.

## 6 Actual simulation results

Figure 2 shows the representative time-of-arrival field from `dx10`. It is read from the actual ELMFIRE output named in the panel, masks nodata, and excludes the two-cell numerical buffer. The 10-m field is the reference-scale spatial result used to interpret mesh-dependent structure spread across the resolution sweep.

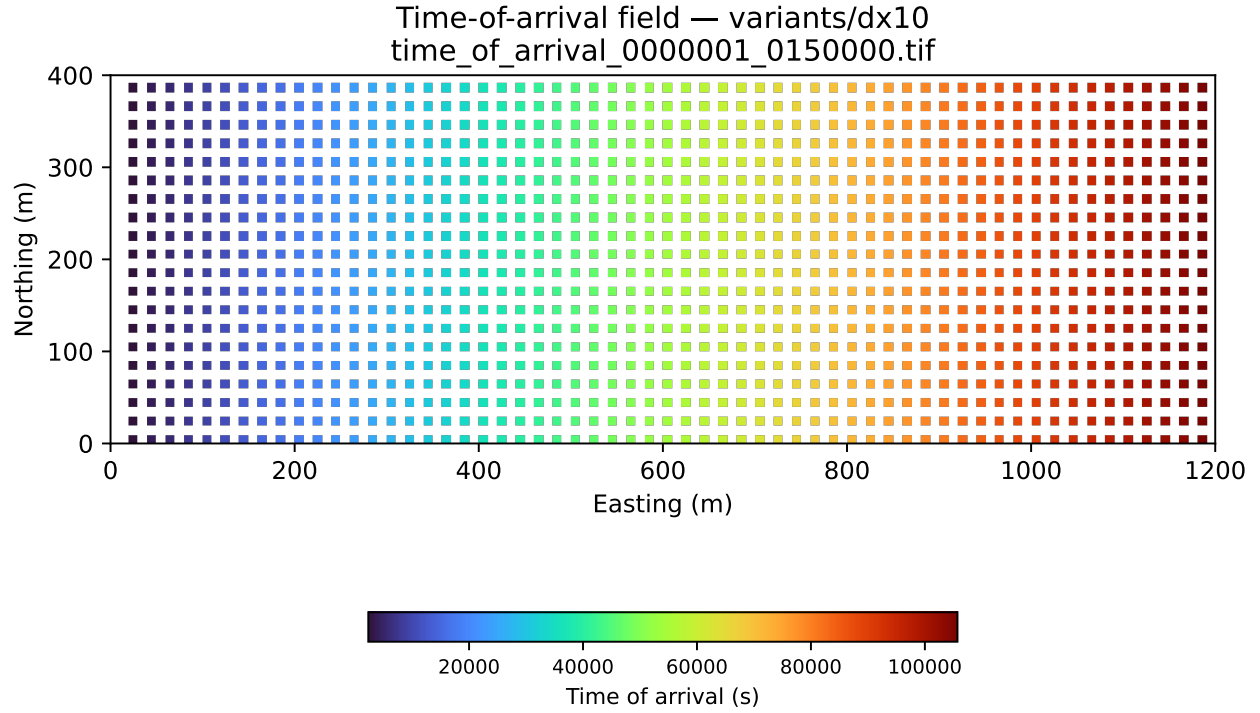


Figure 2: Whole-domain time-of-arrival field from the representative completed ELMFIRE run. This domain-scale result supplements, rather than replaces, the higher-level verification profiles, convergence plots, ensemble statistics, and calculated acceptance metrics below.

All 15 current-fingerprint ELMFIRE processes reached the source success message. The strict terminal-evidence selector accepts 11 of 15 outputs; process exit alone is not used as evidence. Regular terminal dumps are accepted for 1, 3, 5, 7, 10, 20, 30, and 40 m, and the publication-derived no-propagation oracle permits the timestep-aligned stalled-final path for 13, 17, and 25 m. The selected files, configured controls, final times, reconstructed pre-jump times, grid residuals, and rejection reasons are recorded per variant in `outputs/metrics.json`. The overall decision calculated by post-processing is **NOT EVALUABLE**.

The upper row of Figure 3 shows the actual preprocessed geometries. Its lower panel contains fitted ELMFIRE ROS values. Red crosses at zero identify completed simulations with fewer than three ignited downwind columns, for which ROS is deliberately not fitted. The dashed line is  $ROS_{10} = 0.011226 \text{ m s}^{-1}$ , exactly equal to the 10-m ELMFIRE point.

### Two-dimensional spatial-resolution verification

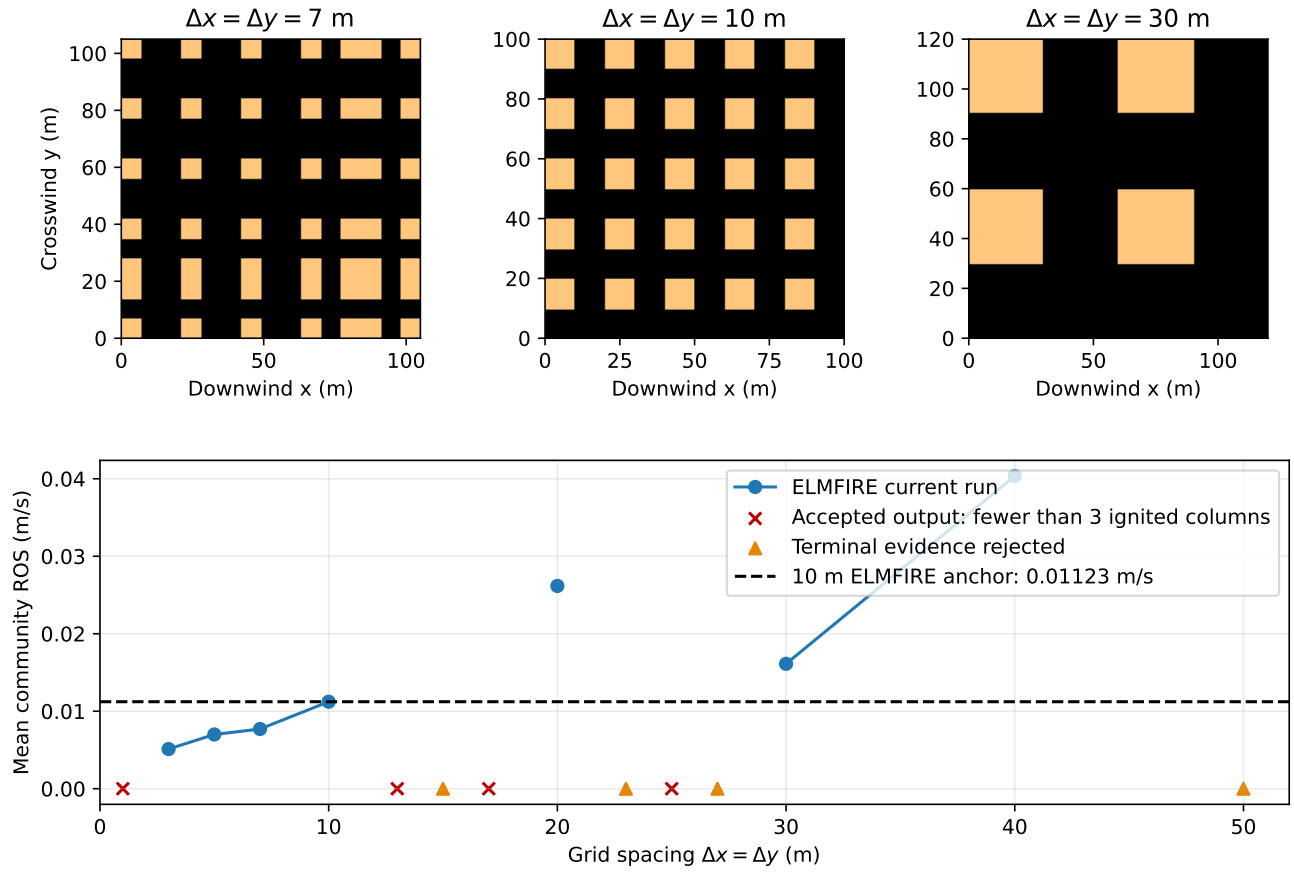


Figure 3: Two-dimensional spatial-resolution diagnostic. Top: mode-resampled WUI geometry at 7, 10, and 30 m. Bottom: mean community ROS from current ELMFIRE TOA outputs; red crosses identify resolutions without enough downwind ignitions for a fit, and the dashed line is the run-derived 10-m anchor. The geometry selection follows the documented comparison configuration [2].

## 7 Calculated metrics and verification decision

Table 3: Calculated verification metrics and component decisions.

| Metric                                 | Acceptance limit         | Calculated | Status        |
|----------------------------------------|--------------------------|------------|---------------|
| Prepared resolution variants           | $\geq 100\%$             | 1          | PASS          |
| dx=10 m footprint error                | $\leq 0.01$              | 0          | PASS          |
| dx=10 m separation error               | $\leq 0.01$              | 0          | PASS          |
| dx=7 m component-area range            | 49 to 196 m <sup>2</sup> | 49 to 196  | PASS          |
| dx=30 m median footprint error         | $\leq 0.01$              | 0          | PASS          |
| Current-fingerprint run completeness   | 15 variants              | 11/15      | NOT EVALUABLE |
| 10-m ELMFIRE ROS anchor                | finite fitted ROS        | N/A        | NOT RUN       |
| ROS-evaluable resolution fraction      | $\geq 100\%$             | N/A        | NOT RUN       |
| ROS mesh-dependence CV                 | $\leq 0.25$              | N/A        | NOT RUN       |
| Maximum ROS deviation from 10-m anchor | $\leq 0.50$              | N/A        | NOT RUN       |

The postprocessor reports **NOT EVALUABLE**. Geometry preparation passes, and all 15 processes executed, but strict terminal evidence is complete for only 11 of 15 variants. The 10-m anchor, ROS-evaluable fraction, coefficient of variation, and maximum anchor deviation therefore remain N/A; calculating them from a selected subset would silently change the predeclared verification population.

## 8 Technical interpretation

The accepted subset contains finite fitted values of approximately  $0.00512 \text{ m s}^{-1}$  (3 m),  $0.00700 \text{ m s}^{-1}$  (5 m),  $0.00770 \text{ m s}^{-1}$  (7 m),  $0.01123 \text{ m s}^{-1}$  (10 m),  $0.02617 \text{ m s}^{-1}$  (20 m),  $0.01612 \text{ m s}^{-1}$  (30 m), and  $0.04036 \text{ m s}^{-1}$  (40 m). These values visibly demonstrate mesh sensitivity, but they are diagnostic observations rather than a complete acceptance calculation.

Four successful processes do not provide admissible terminal evidence. For 15 m, the recorded final time is 152833.592 s and subtracting the configured 3600-s meteorology interval gives 149233.592 s; the timestep-grid residual is 0.302 s. For 23 and 27 m the corresponding reconstructed times are 146507.060 and 148242.909 s, with grid residuals 0.636 and 0.705 s. Those residuals exceed the 0.150-s single-precision-scale tolerance. The 50-m reconstruction is grid-aligned (0.023-s residual), but occurs at 148236.198 s, more than one 2.796-s timestep before the configured stop. The primary publication reports propagation at 15 and 50 m, so neither early stall may be reclassified as independently expected no propagation. Conversely, the publication reports no propagation at 23 and 27 m, but that oracle does not waive the timestep-grid requirement.

The current source also produces finite spread at 3 and 7 m while the published Figure 9 places those resolutions at zero, and it stalls at 15 and 50 m where that figure shows finite spread. The resulting **NOT EVALUABLE** status is therefore an implementation/reference and terminal-time discrepancy, not a missing execution and not a reason to weaken the evidence selector. The runs used executable SHA-256 c225b2e00349aa98128285b3d3ff9b2031d0e20944cd9f1c5ee96595d199a920 built from source commit a2dfbcd72209733c000e5d3431e44723e15efea; the source worktree already contained unrelated local modifications, so the commit and executable digest are both reported for reproducibility.

## 9 Scope and limitations

This case verifies numerical mesh dependence for one idealized periodic WUI array, one wind, and one structural design fire. It does not validate the physical accuracy of any individual ROS value. The 10-m result is an internal run-derived anchor, not experimental validation data. A future resolution-independent structure implementation should be tested with the same geometry and acceptance contract; it should provide all 15 ROS fits and reduce both variation metrics below their limits.

## References

- [1] Y. Qin, *A Physics-Based Eulerian Framework for Modeling Firebrand Showering in Regional-Scale Wildland and Wildland–Urban Interface Fire Simulations*, Ph.D. dissertation, University of Maryland, College Park, 2025.
- [2] Y. Qin et al., “Simulations of firebrand-driven fire spread in landscape-scale Wildland–Urban-Interface and urban conflagration models,” *Fire Safety Journal*, vol. 162, 104686, 2026, doi:10.1016/j.firesaf.2026.104686.